

Multiobjective Optimization

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Web: <https://github.com/brunodeathayde/Multiobjective-optimization/tree/main>

Office Hours: Monday 8:00–9:00 (DEPRO – Block 735)

Classes: Wednesday 10:00–11:40, Friday 10:00–11:40

Course Description

Introduction to multiobjective optimization. Basic foundations. Pareto optimality. A-posteriori preference articulation methods. A-priori preference articulation methods. Interactive preference articulation methods. Performance metrics for evaluating non-dominated solutions. Linear, nonlinear, and combinatorial multiobjective optimization problems. Applications in planning, engineering, and data science. Computational implementation of exact methods and multiobjective metaheuristics.

Rationale

Many real-world problems in engineering, logistics, planning, energy systems, transportation, and data science involve multiple conflicting objectives that must be considered simultaneously. Multiobjective optimization provides a theoretical and methodological framework for modeling and solving such problems, enabling the identification of efficient solutions and supporting decision-making processes. Knowledge of Pareto optimality, preference articulation methods, and modern computational techniques is essential for researchers and practitioners working in complex environments. Furthermore, recent advances in multiobjective metaheuristics have significantly expanded the capability to solve large-scale problems encountered in practice.

Course Objectives

Upon successful completion of this course, students will be able to:

1. Understand the theoretical foundations of multiobjective optimization;
2. Evaluate the quality of solution sets using appropriate performance metrics;
3. Model linear, nonlinear, and combinatorial multiobjective optimization problems;
4. Implement multiobjective optimization algorithms in Python;
5. Develop and analyze solutions for real-world multiobjective optimization applications.

Teaching Methodology

The course will consist of in-person lectures complemented by computational laboratory activities using Python. Theoretical concepts of multiobjective optimization will be presented alongside practical implementations involving modeling, solution methods, and analysis of real-world problems. Students will develop an applied project throughout the semester, culminating in a written report and oral presentation.

Assessment

Assessment will consist of problem sets, computational assignments, a written examination, and an applied project developed throughout the semester. The project will involve modeling and solving a multiobjective optimization problem using methods covered in the course. The final grade will be calculated based on a weighted average of all assessment components. Approval criteria will follow the regulations of the Federal University of Ceara.

References

- COHON, J. L. *Multiobjective Programming and Planning*. Dover Publications, 2013.
- CHANKONG, V.; HAIMES, Y. Y. *Multiobjective Decision Making: Theory and Methodology*. Dover Publications, 2008.
- MIETTINEN, K. *Nonlinear Multiobjective Optimization*. Springer, 2012.
- TOLOO, M.; TALATAHARI, S.; RAHIMI, I. *Multi-Objective Combinatorial Optimization Problems and Solution Methods*. Academic Press, 2022.
- DEB, K. *Multi-Objective Optimization Using Evolutionary Algorithms*. John Wiley & Sons, 2001.
- DEB, K. *Optimization for Engineering Design: Algorithms and Examples*. PHI Learning, 2012.

Course Schedule

- **Lecture 1:** Course introduction: objectives, methodology, references, and grading criteria.
- **Lecture 2:** Basic foundations: optimization concepts, objective functions, and constraints.
- **Lecture 3:** Pareto Optimality I: definitions and simple examples.
- **Lecture 4:** Pareto Optimality II: Pareto frontiers in linear problems.
- **Lecture 5:** Pareto Optimality III: nonlinear problems and practical implications.
- **Lecture 6:** A-priori Methods I: definitions and examples.
- **Lecture 7:** A-priori Methods II: utility functions and weighting methods.
- **Lecture 8:** A-posteriori Methods I: concepts and examples.
- **Lecture 9:** A-posteriori Methods II: solution generation and posterior selection.
- **Lecture 10:** Interactive Methods I: interaction with the decision maker.
- **Lecture 11:** Interactive Methods II: practical examples.
- **Lecture 12:** Python Practice I: determination of efficient (non-dominated) solutions.
- **Lecture 13:** Python Practice II: performance metrics for evaluating non-dominated solutions.

- **Lecture 14:** Python Practice III: metrics for Pareto frontier approximations.
- **Lecture 15:** Python Practice IV: the ε -constraint method.
- **Lecture 16:** Python Practice V: nonlinear multiobjective optimization problems.
- **Lecture 17:** Python Practice VI: combinatorial multiobjective optimization problems.
- **Lecture 18:** Python Practice VII: introduction to multiobjective metaheuristics.
- **Lecture 19:** Python Practice VIII: advanced multiobjective metaheuristics.
- **Lecture 20:** Combinatorial Problems I: introduction.
- **Lecture 21:** Combinatorial Problems II: practical examples.
- **Lecture 22:** Combinatorial Problems III: solution approaches based on metaheuristics.
- **Lecture 23:** Applications in planning: transportation, energy, and resource management.
- **Lecture 24:** Applications in data science: machine learning and model optimization.
- **Lecture 25:** Review I: foundations and Pareto optimality.
- **Lecture 26:** Review II: a-priori, a-posteriori, and interactive methods.
- **Lecture 27:** Review III: Python implementations and performance metrics.
- **Lecture 28:** Written examination.
- **Lecture 29:** Make-up examination.
- **Lecture 30:** Seminar I: student project presentations (Part 1).
- **Lecture 31:** Seminar II: student project presentations (Part 2).
- **Lecture 32:** Seminar III: student project presentations (Part 3).